

Vishal Jeyam

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Education

Master of Science in Computer Science | University of Arkansas

Expected Graduation: May 2027

- **Thesis:** Improving a stock movement prediction model by incorporating news or sentiment data alongside traditional market data

Bachelor of Science in Computer Science | University of Arkansas

August 2021 to May 2025

Work Experience

Research Assistant | Smart Agricultural & Food Engineering Lab

May 2025 to Present

- Operated a Clearpath A200 Husky autonomous mobile robot, integrating ROS-based workflows for sensor fusion, mapping, and navigation as part of an AI-powered environmental monitoring and sanitation system
- Collected and curated large-scale multimodal datasets spanning LiDAR, vision, odometry, and position data across poultry facility environments to support computer vision modeling
- Trained a YOLOv11 segmentation model to classify yellow vs. green plants in real time, achieving 98.1% precision and 92.6% mAP@50 on validation, with 98.2% accuracy on unseen test data, powering the navigation decisions that led to 2nd of 7 teams in both the autonomous navigation and identification categories

Undergraduate Research Intern | Washington State University

June 2024 to August 2024

- Created a real-time crop health and yield prediction pipeline for a WSU wheat breeding program (REEU), using a provided dataset of 7,000+ paired NIR and RGB multispectral images
- Fine-tuned a YOLOv8 model to detect and segment wheat field plots, achieving 82% detection accuracy and 84% segmentation accuracy, to isolate crop regions for downstream analysis
- Computed vegetation indices (Soil Crop Index, Green Normalized Difference Vegetation Index) within segmented plots to predict wheat health and yield, reducing the cost and manual labor of traditional breeding-program assessment

Software Engineer Intern | Arkansas Blue Cross and Blue Shield

June 2023 to August 2023

- Built a proof-of-concept Kafka consumer client in Python to receive real-time data from an existing producer pipeline, validating end-to-end delivery by confirming a test dataset of team member names published by the producer was correctly received by the consumer
- Developed a proof-of-concept Elasticsearch ingestion script to repeatedly pull data from one or more PostgreSQL databases and consolidate it into a single unified database, enabling cross-team access to previously siloed data
- Contributed to the Integration Hub team's pipeline lifecycle, spanning stream ingestion with Kafka to search indexing with Elasticsearch, across distributed systems

Projects

Sports Odd Pipeline

February 2026

- Designed a modular ETL pipeline in Python to ingest real-time sports betting odds from major U.S. markets via the Odds API and game results from the ESPN API, storing structured data in a SQLite database
- Engineered sports market efficiency analytics including close line value, calibration modeling, and best-market frequency tracking, finding pregame odds correctly predicted game outcomes 70% of the time versus a 50/50 baseline
- Developed an interactive Streamlit dashboard with strategy simulation metrics (ROI, drawdown, profit factor) and one-click ETL refresh, supporting both end-user analytics and admin-level pipeline monitoring

Skills

- Programming Languages: Python, SQL, C++, HTML/CSS
- Frameworks & Libraries: PyTorch, Scikit-learn, React
- Data & Tools: Kafka, Elasticsearch, Spark, Git/GitHub, Tableau, Pandas, NumPy
- Cloud & DevOps: Docker, Azure